

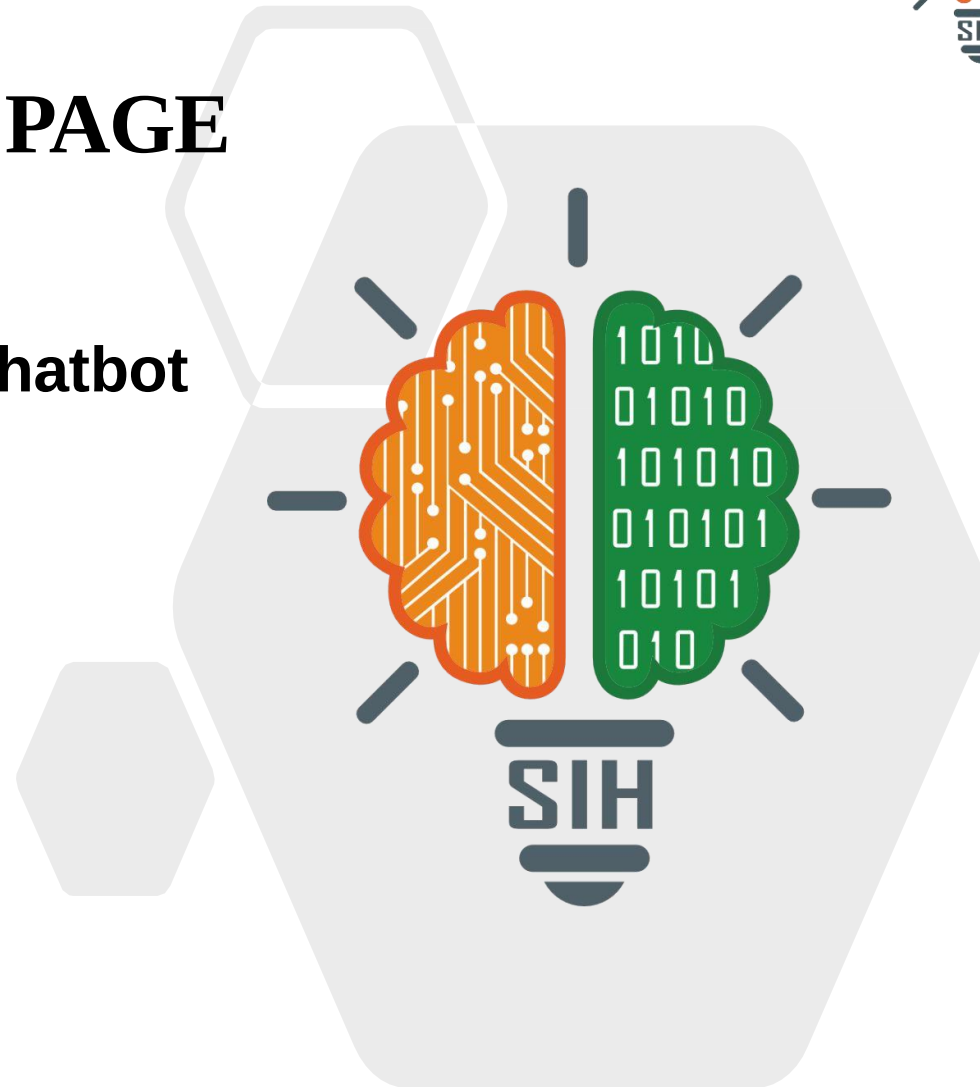
SMART INDIA HACKATHON 2024



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TITLE PAGE

- Problem Statement ID – 1648
- Problem Statement Title- Online Chatbot Based Ticketing System
- Theme- Travel And Tourism
- PS Category- Software
- Team ID- 2698
- Team Name- Flow Dynamics



IDEA / SOLUTION :

Online Chatbot Based Ticketing System

- ❖ Our solution involves creating a **conversational AI-based** chatbot for the National Council of Science Museums (NCSM).
- ❖ Our AI-driven chatbot will handle ticket booking, customer support, FAQs, and provide detailed museum information, offering visitors real-time assistance and
- ❖ **Augmented Reality** (AR)will provide exhibit content, allowing visitors to explore more activities.
- ❖ The chatbot will offer **multilingual and voice support**, ensuring accessibility for a diverse range of visitors, including international basis.
- ❖ Visitors can book tickets in real-time and receive **personalized email** updates about upcoming events, keeping them informed and engaged with the museum.

Problem Resolution :

- ❖ Unified Booking System
- ❖ Enhanced Visitor Experience
- ❖ Real-Time Information & Updates
- ❖ Rewards and Discounts
- ❖ Multilingual and Voice Support
- ❖ Handles double bookings and payment failure.

Unique Value Propositions (UVP) :

- ❖ Personalized Recommendations
- ❖ AR Integration
- ❖ Targeted Email Campaigns
- ❖ Advanced Chatbot Capabilities
- ❖ Dynamic Pricing & Discounts

Language Used: Python

Frontend Infrastructure:

React.js , SASS, Tailwind.CSS, Node.js - Core technologies used for developing the Web application development.

Backend Infrastructure:

Langchain , NLP ,RAG -Framework used for developing the Chatbot.

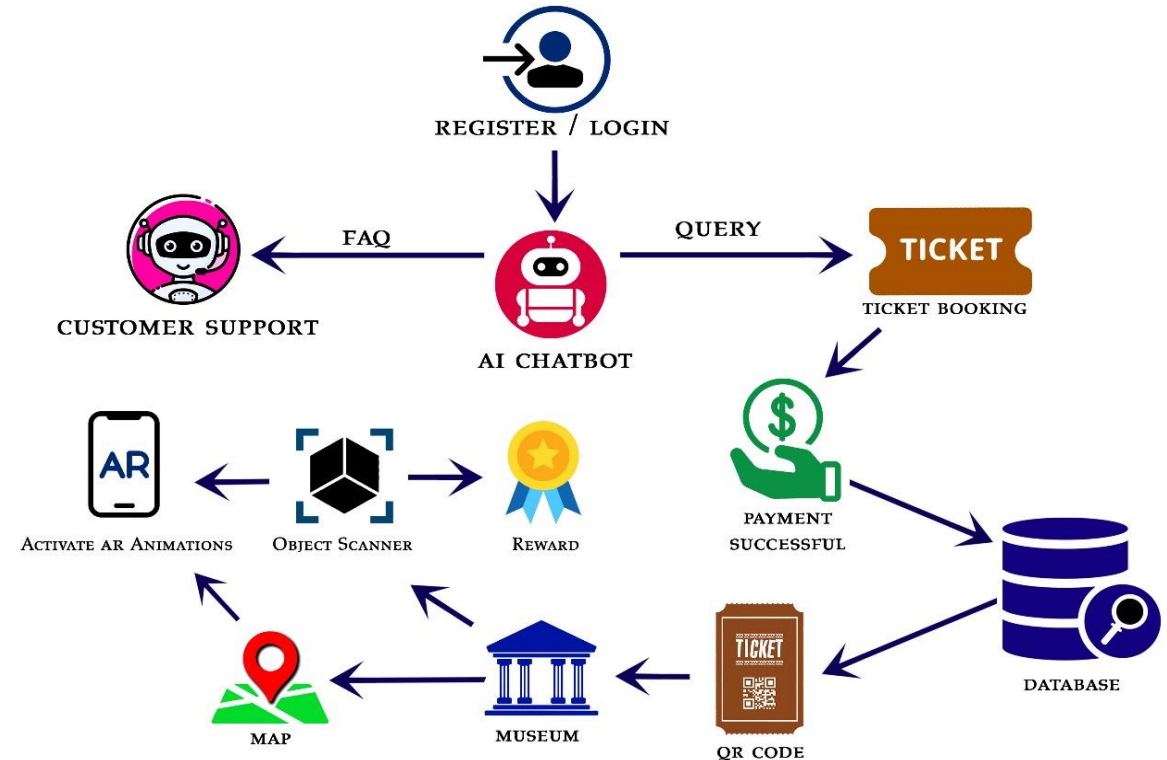
Cloud Management:

MongoDB Atlas, MySql- Used for managing the cloud database and depending on the data structure and scalability needs.

AR Development:

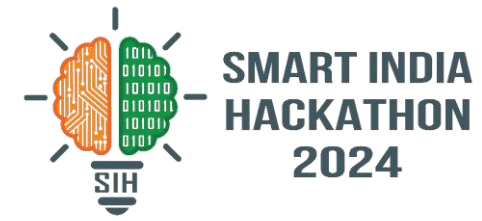
Blender ,unity - used for implementing Augmented reality.

PROCESS FLOW ARCHITECTURE



Project Status: 70% project built completed and further build is on progress. Testing and validation process are next to be undergone.

FEASIBILITY AND VIABILITY



- **Analysis of the feasibility of the idea**
 - **Technical**- Chatbot Development , Integration with existing systems , Scalability
 - **Financial**- Development Cost , Ongoing Expenses , Revenue Potential
 - **Operational**- Staff Training , Content Creation , User Experience
- **Potential challenges and risks**
 - Over-reliance on technology and loss of human touch
 - Insufficient Funding and Resources
 - Dependence on third party services and API's
- **Strategies for overcoming these challenges**
 - **Method**-Conduct regular data analysis to monitor technical performance and user behavior
 - **Strategies**-Develop a user experience strategy (Personalization and accessibility)
 - **Principle**-Prioritize user needs and expectation in all aspects of the chatbot system

- **Potential impact on the target audience**

- ❖ **Positive:**

- Streamlined Ticket Booking
 - Automated Customer Support
 - Easy Deployment Across Museums
 - AR Exhibit Growth
 - Reward Points System
 - Increased visitor satisfaction

- ❖ **Negative:**

- Information Overload
 - Equity and access

- **Benefits of the solution**

- ❖ **Social:**

- Real-time feedback
 - Social Media Integration
 - Increased Engagement
 - Educational Value

- ❖ **Economic:**

- Cost savings
 - Boosted Tourism and Local Economy
 - Merchandising and Donations
 - Exhibit Packages

Research Studies

- A Study of User Experience by the Museums and the Web conference (2019)
- The Impact of Chatbots on Museum Visitor Engagement by the Journal of Museum Studies (2020)
- Evaluating the Effectiveness of Chatbots in Cultural Heritage Institutions" by the International Journal of Heritage Studies (2020)

References

- <https://ncsm.gov.in/> National Council of Science Museums (NCSM)
- <https://nationalmuseumindia.gov.in/en/online-ticket-booking> National Museum of New Delhi